



## Andreia Mendes

✉ Email address: [andreia.fer.mendes@gmail.com](mailto:andreia.fer.mendes@gmail.com) ☎ Phone: (+351) 927416449

🌐 Website: <https://gatinha96.github.io/portfolio/>

**Gender:** Female **Date of birth:** 16/12/1998 **Place of birth:** Portugal **Nationality:** Portuguese

### ABOUT MYSELF

I first came into contact with the world of programming during a summer internship, and likening it to puzzle solving which I've always loved, my career choice became clear. I then decided to study programming as part of a game development university course, since I also love videogames, mostly using C# and C++, on platforms such as Unity and Monogame framework, among others, and participated in gamejams. (A coding challenge where we have 48h to create a game with a team according to a theme)

I acquired a scholarship as part of my master's degree, and there I developed an application (a serious game) for the therapy and/or diagnosis of schizophrenia in VR. This project became my dissertation focus (more details are included in the "Projects" section of this website).

Later on, I acquired a job at the CCG/ZGDV research center, becoming part of the immersive environments department, which includes AR and VR projects. With the nature of a research center, I've often learnt new techniques or worked with new technologies, software and hardware, which I've greatly enjoyed.

I have an English Cambridge diploma, and I communicate in English comfortably on a daily basis, engaging with people around the world on a discord server which I assist with as part of administration, with hundreds of members.

### EDUCATION & TRAINING

[ 2015 ]

#### Cambridge certificate CAE level (C1)

*Cambridge*

[ 01/09/2007 - 31/07/2016 ]

#### Ciências e Tecnologias (Sciences and Technologies)

*Didáxis Vale S. Cosme* <https://www.didaxis.pt/regular/>

**Address:** Av. Tibães 1199, Vale de São Cosme 4770-568, Vila Nova de Famalicão (Portugal)

[ 19/09/2016 - 31/07/2019 ]

#### Licenciatura em Engenharia e Desenvolvimento de Jogos Digitais (Degree in Engineering and Development of Digital Games)

*Instituto Politécnico do Cávado e Ave (IPCA)* [www.ipca.pt](http://www.ipca.pt)

**Address:** Campus do IPCA - Lugar do Aldão, 4750-810 Vila Frescainha (São Martinho) 4750-810, Barcelos (Portugal)

[ 19/09/2019 - 31/08/2022 ]

#### Mestrado em Engenharia e Desenvolvimento de Jogos Digitais (Masters in Engineering and Development of Digital Games)

*Instituto Politécnico do Cávado e Ave (IPCA)* [www.ipca.pt](http://www.ipca.pt)

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Martinho)4750-810, Barcelos (Portugal)

## WORK EXPERIENCE

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### **Bosch**

**City:** Braga | **Country:** Portugal

[ 01/07/2016 - 31/08/2016 ]

### **Engineering assistant**

First work experience in an industrial environment through a summer internship at Bosch. Assisted with data management in Excel while gaining exposure to the operations of a large company. This experience also marked my first introduction to programming through interactions with the IT department, inspiring me to pursue a degree in the field.

### **Instituto Politécnico do Cávado e Ave (IPCA)**

**City:** Barcelos | **Country:** Portugal

[ 01/03/2021 - 31/01/2023 ]

### **Application programmer**

Work internship facilitated by IPCA in collaboration with a medical team, focused on developing a VR serious game for schizophrenia therapy and/or diagnosis. This project later became the subject of my master's thesis.

### **CCG/ZGDV Intitute** <https://www.ccg.pt/>

**City:** Guimarães | **Country:** Portugal | **Name of unit or department:** Computer Vision, Interaction and Graphics (CVIG) | **Business/sector:** Professional, scientific and technical activities

[ 01/10/2023 - Current ]

### **Software developer**

Development of AR/VR solutions and conducting research for companies. Finding creative solutions to complex problems and pushing the boundaries of AR/VR technology to drive innovation. Tested applications on devices such as smartphones, Oculus Quest 2 & 3, and HoloLens. Experienced leveraging compute shaders to optimize performance and enhance immersive experiences.

## LANGUAGE SKILLS

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**Mother tongue(s):** Portuguese

**Other language(s):**

**English**

**LISTENING:** C1 **READING:** C1 **WRITING:** C1

**SPOKEN PRODUCTION:** C1 **SPOKEN INTERACTION:** C1

## SKILLS

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### **Languages**

C# | C++ | C | SQL | GLSL/HLSL | Swift-Xcode (Basic) | ShaderLab | LaTeX | JavaScript  
| compute shaders | shadergraph

### **Tools**

3DS Max | GitHub Desktop | Maya | Figma | Blender | Overleaf | Microsoft: Microsoft Excel, Microsoft PowerPoint, Microsoft Word, Microsoft Outlook

### **Domains**

AR | XR | VR | Frontend | Backend | game development | modeling | mobile development | multithreading

## Frameworks

MRTK | AR Core | XR Toolkit | Visual Studio | Visual Studio Code | Monogame framework | OVR | Unity | Windows Forms | SQL Server | Git

## PROJECTS

[ 01/05/2026 - 18/05/2026 ]

### Portfolio Website

Creation of a website to host my portfolio and general information, using Javascript, HTML and React.js.

Starting from a premade website skeleton, the site was further refined and adapted to my needs.

Additional information not present on this CV is included in the website, as well as additional photos, videos and projects.

**Link:** <https://gatinha96.github.io/portfolio/>

[ 30/11/2023 - Current ]

### Cloth and clothes simulation - Texp@ct WP3 - PPS13

Texp@ct is a response to the challenge of digital transition and adaptation to Industry 4.0 in the textile and clothing sector.

The Pact for Innovation in the Textile and Clothing Industry will develop a set of 26 digital solutions and technologies, for different areas of work, namely robotics, product engineering, industrial processes, artificial intelligence, smart textiles, data sharing and digital skills.

With this pact, it is intended that there will be a significant boost in the adoption of innovative digital technologies and solutions, making the sector more attractive for current and future global markets and for human resources, more resilient, more sustainable and more sophisticated.

This project is financed by the Recovery Plan and Resilience.

For this project, I conducted research and development of a tool capable of simulating cloth and clothing realistically in real time on a human model. Realistic cloth simulation is challenging due to the complex nature of fabric behavior, which involves factors such as elasticity, friction, and interaction with environmental forces like wind and gravity. Additionally, simulating the deformation of these complex structures requires precise calculations of how the fabric moves and responds to the underlying human model, which adds significant computational load.

The most challenging aspect of this project is optimization, as physics simulation is resource-intensive, and realistic cloth physics involves simulating complex deformable structures. Furthermore, since this tool is intended to run in a web environment, available computational resources are further limited.

A careful balance of performance and quality has been achieved, with quality was further enhanced with the help of lightweight shaders, and performance boosted by the use of compute shaders.

**Link:** <https://www.youtube.com/watch?v=XMux8fM3f20>

[ 01/10/2023 - Current ]

### Virtual Museum for HFPT - Health from Portugal

The Health from Portugal (HfPT) Agenda aims at positioning Portugal as a world reference hub in the design, development and production of advanced solutions for the health markets, based on innovation and technology.

The HfPT Agenda is designed to provide an integrated response to the challenges facing the health sector, taking advantage of the emerging opportunities surrounding the digital and green transition and economic resilience. In general terms, the Mobilizing Agenda aims to position Portugal as a reference hub in the design and development of healthcare products and services that stand out for their degree of innovation and incorporation of technology.

The Smart Health Solutions Vertical is designed to foster a new paradigm for the operation and management of healthcare systems, geared towards the collection and analysis of data generated by the devices to be developed. These innovations are expected to stimulate the definition of new digital health models based on digital and patient-centered, with a focus on reducing costs and improving the quality of the services provided.

WP5 - Digital consultation and telemedicine | A5.3.4: Virtual museum

Capacity building for the growth and sophistication of the clinical studies on the country. Web app for visitors of FMUC's Museum of Anatomical Pathology to have access to additional info and digital twins of exposed pieces in Augmented Reality. The aim is to allow for easier access for medicine students to visit and inspect the museum and its items.

**Link:** <https://www.youtube.com/watch?v=fMbwnmRuo-U>

[ 30/11/2023 - Current ]

### **Development of a realistic and customizable human avatar**

Developed realistic human avatars that users can customize through sliders for various physiological aspects, such as arm and leg length, height, body weight, and more. Each aspect is represented internally as model information, defining the minimum and maximum values for each slider, which are then blended based on user input.

The most challenging aspect of this project was optimization, as it needed to run efficiently in a web environment. The initial algorithm iteration was executed on the CPU, but modern techniques shifted the processing to the GPU to utilize its idle capacity and free up the CPU for other calculations.

By implementing compute shaders and multithreading, we significantly accelerated the blending process from just under 20 seconds to a fraction of a second. This optimization enabled real-time customization of avatars without lag.

The customizable avatar component is part of a larger project (Texp@ct WP3: PPS13) and has been successfully integrated into the overall project.

**Link:** <https://www.youtube.com/watch?v=XMux8fM3f20>

[ 30/11/2023 - Current ]

### **Prediction of thread/sample aspect**

Developed an application capable of estimating the final appearance of blended thread materials. For this project, I created a customizable shader that allows users to define and create the thread materials they want to blend, which will be used by an algorithm that blends these materials based on varying weights (how much of each material does the blend mix) to produce a final product visualization. Key aspects such as smoothness, reflection, and color were carefully considered to ensure accurate representation of the blended materials.

After this groundwork, the project has been moved to other teams to be further refined.

[ 30/11/2023 - 01/01/2025 ]

### **Development of a train simulator in XR**

Conducted research and development of a prototype for a realistic train simulator, featuring real-world tracks and trains. The simulator supports various levels of immersion, from traditional on-screen simulation to full surround and VR experiences, with control options ranging from virtual interfaces to the recreation of actual train controls.

Research indicated that using OpenRails as a foundation is viable due to its open-source nature, which already includes physics for different types of trains, train systems, and track systems, as well as tools for track building. Additionally, I developed realistic shaders to simulate weather effects, such as fog, rain, and snow.

Unfortunately this project did not get the green light to be further developed.

[ 01/03/2021 - 31/01/2023 ]

### **Development of a VR game for the therapy and/or diagnosis of Schizophrenia**

Schizophrenia is a severe mental disorder that distorts a person's perception of reality, often necessitating lifelong treatment. In recent years, innovative treatment methods, including serious games and VR technologies, have emerged. This project aims to develop a VR tool to assist in the therapy of schizophrenia patients, tested under medical supervision to evaluate its viability as an alternative to traditional therapy due to its ease of use and convenience.

Collaborating with a medical team, we designed game tasks while being mindful of the environments and exercises presented to patients. For example, to avoid triggering paranoia—which is common among schizophrenia patients—we deliberately excluded mirrors from the environment, as patients may misinterpret their reflections as strangers.

Initially, we aimed for the game to be compatible with multiple VR systems. However, this proved challenging due to the lack of universal software across different platforms. As a result, we shifted focus to creating the most immersive experience possible, ultimately choosing the Oculus Quest 2, a wireless system that supports hand tracking and is among the most accessible VR options.

We developed the game using the Unity platform, which is well-suited for Oculus Quest 2 development and offers ready-to-use frameworks. During development, we explored three frameworks: XR Toolkit, OVR Toolkit, and Mixed Reality Toolkit (MRTK). Initially, we used both XR and OVR, with XR supporting broader VR system integration, while OVR was used for hand tracking. Eventually, we transitioned from XR to MRTK for compatibility with HTC Vive, as XR was not suitable.

Ultimately, we decided to focus solely on the Oculus Quest 2 and hand tracking to ensure an optimal user experience. I was responsible for the majority of the game's development, including UI, game flow, mechanics, and VR integration. This project was a significant challenge for me, as I had no prior experience in VR development, but I embraced the opportunity to learn and grow.

**Link:**

[https://www.researchgate.net/publication/366353663\\_Sensors\\_Transducers\\_Development\\_of\\_a\\_Virtual\\_Reality\\_Tool\\_for\\_Therapy\\_and\\_Diagnosis\\_of\\_Schizophrenia](https://www.researchgate.net/publication/366353663_Sensors_Transducers_Development_of_a_Virtual_Reality_Tool_for_Therapy_and_Diagnosis_of_Schizophrenia)

[ 30/11/2019 ] **Game engine**

For this project, I built my own game engine using the MonoGame framework instead of relying on a pre-existing engine like Unity. My goal was to create a highly optimized engine that maximized performance.

To evaluate its efficiency, I implemented a test scenario rendering a spaceship model. The objective was to render as many instances as possible while maintaining a stable frame rate. The ships moved in a set direction and were destroyed upon reaching a certain distance.

To optimize performance, I employed techniques such as object pooling, optimized data structures, and rendering optimizations like culling unnecessary triangles. As a result, my modest computer, equipped with a GeForce 965M graphics card, successfully rendered over ten thousand ships while maintaining 60+ FPS for an entire hour.

This project provided deep insight into optimization and demonstrated the power of efficient code. Ironically, the same computer now struggles with certain unoptimized software, highlighting the importance of performance tuning.

Developed using Visual Studio and MonoGame, and programmed in C#.

**Link:** <https://github.com/gatinha96/Motores-de-jogos>

[ 2019 ] **Randomly generated terrain using multithreading for a game**

For this project, I combined concepts from two university courses—Artificial Intelligence and Optimization Techniques—to develop a terrain generation system using noise algorithms and multithreading. The goal was to efficiently create a unique, playable

environment each time it was generated.

Terrain generation was achieved using Perlin noise, manipulating a plane mesh to form dynamic landscapes. A PNG image was also generated, assigning colors based on biome, height, neighboring biomes, and a chaos factor to introduce diversity. To enhance efficiency, I implemented multithreading, reducing generation time from approximately 2 minutes to 30 seconds—making calculations nearly four times faster.

Later, I expanded the project by incorporating concepts from my Game Projects course. The generated terrain and multithreading optimizations became the foundation for a full game, featuring creature and object generation, inventory systems, player movement, and cave generation using a recursive backtracker algorithm for labyrinth creation.

This project was developed in Unity and programmed in C#.

[ 2018 ] **SQL server integration with Windows Forms**

Developed a SQL Server database and a complementary Windows Forms application for managing a shop-themed system, including sales data, customer information, and product details. The application featured pre-built queries, such as adding new sales, and allowed users to send custom queries to the connected SQL Server. Query results were displayed in tables, enabling tasks like filtering product sales above a certain price. Built using SQL Server, Visual Studio, and Windows Forms, and programmed in C# and SQL.

[ 30/11/2017 ] **AR Mobile Game**

Completed a course in AR development, where I learned to create AR games using ARCore, a software system that detects surfaces and markers via camera systems. However, due to hardware incompatibility, I explored an alternative approach by leveraging the phone's gyroscope to simulate an AR experience.

In my project, players had to physically turn their phones to locate and interact with creatures appearing around them. While the camera was used for cosmetic purposes, the core gameplay relied on gyroscope-based orientation. The game featured increasing difficulty, with enemies becoming more numerous and challenging over time.

This creative approach, diverging from taught AR techniques, was praised by my teacher. The project was developed using Unity and programmed in C#.

**Link:** <https://www.youtube.com/watch?v=3jKHHeKtZXw>